Estimating Semiparametric and Nonparametric Fixed Effects Panel Data Models with `mgcv`

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Abstract

This paper provides a practical guide to estimating semiparametric and nonparametric fixed-effects panel models using `mgcv` in R. It covers unit indicators, first differencing, penalized unit effects, smooth specification, and cluster-robust inference. Monte Carlo experiments compare `mgcv::bam` with linear and fixed-series spline estimators. Penalized splines adapt to functional complexity and estimate functions accurately in these designs. With unit indicators or first differences, penalty-adjusted cluster-robust inference yields near-nominal size for parametric coefficients and pointwise coverage for unknown functions. Uniform bands modestly under-cover; model-based covariance inflation generally reduces undercoverage but can be conservative.

Keywords: semiparametric regression, nonparametric regression, panel data, fixed effects, generalized additive models, splines

JEL codes: C14, C15, C23, C87

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1 Introduction

Empirical work in economics frequently uses panel data to account for unobserved unit-level heterogeneity through fixed effects. At the same time, many applied questions call for flexible functional forms. For example, Borri et al. (2022) estimate a nonlinear investment curve, while Baglan and Yoldas (2014) estimate an inflation–growth relationship semi-parametrically. Such empirical applications remain relatively rare, however, despite a well-developed theoretical literature on semiparametric and nonparametric panel data models.

A central practical obstacle is the need to handle fixed effects while also choosing basis dimension, series length, bandwidths, or smoothing penalties. Nonparametric methods that are popular in cross-sectional settings, such as kernel or nearest-neighbor estimators, do not naturally accommodate additive fixed effects. Series-based approaches can incorporate fixed effects more directly, but they require the researcher to specify basis functions, choose interaction terms, and select the number of series terms.

This paper provides a practical guide to estimating semiparametric and nonparametric fixed effects panel data models using penalized splines. The approach connects three literatures. The first is the econometric literature on series estimation for semiparametric panel data models with fixed effects (Baltagi and Li, 2002; An et al., 2016). The second is the broader econometric literature on sieve and penalized sieve estimation and inference for nonparametric and semiparametric models (Ai and Chen, 2003; Chen, 2007; Chen and Pouzo, 2009, 2012, 2015). The third is the statistical literature on generalized additive models, smoothing parameter selection, and penalized regression splines (Wood, 2003, 2011, 2017).

Conceptually, the proposed approach is grounded in the econometric theory of series, sieve, and penalized sieve estimation. Computationally, it uses the fast and stable implementations developed in the generalized additive model literature, making flexible estimation practical even in panel settings with high-dimensional fixed effects.

This paper is most closely related to Pütz and Kneib (2018), whose estimator is implemented in the accompanying R package `pamfe` (Pütz, 2017). They develop a first-difference penalized-spline estimator and simultaneous confidence bands for additive and partially linear fixed-effects panel models. Their baseline framework assumes homoskedastic, serially independent Gaussian errors in levels and uses generalized least squares to account for the dependence induced by first differencing. The implementation considered here accommodates additive models as well as fully nonparametric multivariate, semiparametric partially linear, and semiparametric varying-coefficient specifications. It handles unit heterogeneity through unit indicators or first differences, with penalized unit effects included as a regularized alternative. For inference, I use a cluster-robust covariance estimator that permits heteroskedasticity and general serial dependence within units without requiring the researcher to specify the within-unit covariance structure.

The paper gives concrete implementation guidance in R using the `mgcv` package (Wood, 2025). I first describe the class of fully nonparametric, partially linear, and varying-coefficient panel models covered by the approach. I then show how the resulting penalized-spline estimators can be implemented using unit indicators or first differences, how a penalized unit-effect specification can be used as a regularized alternative, and how to construct penalty-adjusted cluster-robust standard errors and confidence bands from the fitted `bam` object. Monte Carlo simulations evaluate estimation accuracy for the unknown function, empirical size and power for finite-dimensional parameters, and coverage of confidence bands for the nonparametric component. An empirical application illustrates the proposed approach using employment equations estimated from the publicly available `EmplUK` panel distributed with the R package `plm`.

2 Models

The proposed approach can handle a variety of fixed effects panel data models. One example is the fully nonparametric model

$$Y_{it} = g(X_{it}) + \mu_i + \varepsilon_{it}, \qquad E[\varepsilon_{it} \mid X_{i1}, \ldots, X_{iT}, \mu_i] = 0, \tag{2.1}$$

where X_{it} may be scalar or vector-valued, $g(\cdot)$ is unknown, and μ_i is a unit fixed effect.

Another example is the partially linear model

$$Y_{it} = X_{1,it}'\beta + g(X_{2,it}) + \mu_i + \varepsilon_{it}, \qquad E[\varepsilon_{it} \mid X_{i1}, \dots, X_{iT}, \mu_i] = 0, \tag{2.2}$$

where $X_{it} = (X_{1,it}', X_{2,it}')'$, β is an unknown finite-dimensional parameter, and $g(\cdot)$ is an unknown function.

A third example is a varying-coefficient model,

$$Y_{it} = \beta_0(X_{2,it}) + X_{1,it}'\beta_1(X_{2,it}) + \mu_i + \varepsilon_{it}, \qquad E[\varepsilon_{it} \mid X_{i1}, \dots, X_{iT}, \mu_i] = 0, \tag{2.3}$$

where $\beta_0(\cdot)$ is an unknown intercept function and $\beta_1(\cdot)$ is an unknown smooth coefficient function, or vector of coefficient functions when $X_{1,it}$ is vector-valued.

In these additive fixed-effects models, the level of the unknown smooth is not separately identified from the intercept and unit effects. The identified object is a normalized version of the function. For population statements one can write $g^c(x) = g(x) - E[g(X_{2,it})]$, with the obvious modification in the fully nonparametric model. The tables below use the operational grid analog

$$g_G^c(x_j) = g(x_j) - \frac{1}{J}\sum_{\ell=1}^{J} g(x_\ell), \qquad j = 1, \dots, J,$$

on the evaluation grid $G = \{x_1, \dots, x_J\}$. For RMSE calculations, the same normalization is applied to an independent test set. For first-differenced estimators, the data identify smooth contrasts $g(X_{2,it}) - g(X_{2,i,t-1})$; reported level curves are reconstructed in the fitted basis and then normalized using the same grid-centering convention.

I assume throughout that there is sufficient within-unit variation and connected covariate support to distinguish changes in the smooth from the unit effects. The partially linear and varying-coefficient specifications additionally require the usual rank and support conditions separating their parametric and nonparametric components.

3 Estimation: Penalized Objective

This section describes the penalized objective that underlies the estimators used in this paper and the role of fast restricted maximum likelihood (fREML) in selecting the penalty weights. A Gaussian identity-link `gam` or `bam` fit can be viewed as a penalized least squares problem with the amount of penalization chosen automatically. After each smooth term has been represented by a finite set of basis functions, the model is linear in its unknown coefficients. Let R denote the resulting `mgcv` model matrix, including parametric regressors, spline basis columns, and any factor indicators, and let θ collect all coefficients. For a fixed vector of smoothing parameters $\lambda = (\lambda_1, \dots, \lambda_J)'$, `mgcv` estimates

$$
\begin{aligned}
\hat{\theta}_\lambda &= \underset{\theta}{\operatorname{argmin}}\, Q_\lambda(\theta), \\
Q_\lambda(\theta) &= \sum_{i=1}^{n} \sum_{t=1}^{T} \left(Y_{it} - R_{it}'\theta\right)^2 + \sum_{j=1}^{J} \lambda_j \theta' S_j \theta \\
&= \|Y - R\theta\|^2 + \theta' S_\lambda \theta, \qquad S_\lambda = \sum_{j=1}^{J} \lambda_j S_j.
\end{aligned}
\tag{3.1}
$$

The matrices S_j are known positive semidefinite penalty matrices constructed from the chosen smooth bases; each S_j has nonzero entries only for the coefficients belonging to the corresponding smooth term. This is directly analogous to the familiar ridge-type objective

$$
\min_{\beta_0,\dots,\beta_k} \sum_i \left(Y_i - \beta_0 - \sum_{j=1}^{k} X_{ij}\beta_j \right)^2 + \lambda \sum_{j=1}^{k} \beta_j^2.
$$

Here the intercept is left unpenalized, while larger values of λ shrink the slope coefficients toward zero. In the spline case, however, the row R_i contains the usual regressors, spline basis functions, fixed-effect indicators, and any other columns generated by the model formula, rather than only the raw covariates X_i.

For example, in the partially linear model in equation (2.2), the specifications used below set `select = TRUE`, so a schematic block representation is

$$R = [X_1 \;\; B(X_2) \;\; D], \qquad \theta = (\beta', \gamma', \mu')', \qquad S_\lambda = \text{blockdiag}\Big(0, \lambda_1 \widetilde{S} + \lambda_2 \widetilde{S}_0, 0\Big),$$

where $B(X_2)$ contains the spline basis terms, D contains the unit indicators, $\widetilde{S}$ penalizes the wiggliness of the spline coefficients γ, and $\widetilde{S}_0$ is an additional penalty that shrinks the otherwise unpenalized components toward zero, allowing the entire smooth term to be effectively removed from the model. The smoothing parameters λ_1 and λ_2 are selected by fREML. The parametric regressors and unit effects are unpenalized. In practice, `mgcv` imposes an identifiability constraint on the smooth and reparameterizes the basis and penalties, so the corresponding columns and penalty blocks in the fitted object are transformed versions of $B(X_2)$, $\widetilde{S}$, and $\widetilde{S}_0$.

For each smooth, the user-specified basis dimension K_n (the argument `k` in `mgcv`) determines the number of available basis functions and hence the finite-dimensional approximation space. It does not generally equal the effective degrees of freedom of the fitted smooth.

For common cubic and thin-plate spline bases, the quadratic form $\theta' S_j \theta$ is a finite-dimensional measure of wiggliness, such as an integrated squared derivative. Thus λ_j controls how costly wiggliness is for the jth smooth. A large λ_j pushes the smooth toward the null space of its penalty, such as a low-order polynomial component; with `select = TRUE`, `mgcv` adds an additional shrinkage penalty on this null space, so that a smooth can be penalized all the way toward zero. Ordinary parametric regressors and factor fixed effects are not penalized unless an explicit parametric penalty is supplied. By contrast, a term such as `s(unit, bs = "re")` uses a ridge-type penalty on the unit effects, corresponding to a Gaussian random-effect representation.

The label `bs = "re"` refers to a computational random-effect representation of a ridge penalty on the unit indicators, not to the classical random-effects assumption that the unit effect is uncorrelated with the regressors. Conditional on the smoothing parameter, the estimator is penalized least squares with a quadratic penalty on the unit effects. Because this specification shrinks the unit effects toward zero, however, the resulting estimator differs numerically from including unpenalized factor fixed effects. When the unit effects are correlated with the regressors and T is fixed, this shrinkage can transmit part of that correlation into the estimated smooth or finite-dimensional coefficient. I therefore treat `s(unit, bs = "re")` as a predictive or regularized alternative, not as a substitute for the fixed-effects estimand delivered by unpenalized unit indicators or first differencing.

It is important to distinguish the coefficient objective in equation (3.1) from the role of `method = "REML"` or `method = "fREML"`. Conditional on λ, the coefficient estimates solve the penalized least squares problem above; REML or fREML is then used to choose λ and the scale parameter. In the Gaussian case, this can be understood by treating the penalized spline coefficients as Gaussian random effects with precision proportional to the smoothing parameters. For a trial value of λ, `mgcv` solves the penalized coefficient problem and evaluates a restricted marginal likelihood criterion, with the scale parameter profiled or updated. The outer smoothing-parameter step then searches over λ. Schematically, up to constants and rank adjustments, the negative REML criterion has the form

$$\mathcal{V}_{\mathrm{REML}}(\lambda, \sigma^2) = \frac{1}{2\sigma^2} Q_\lambda(\hat{\theta}_\lambda) + \frac{1}{2}\log|R'R + S_\lambda| - \frac{1}{2}\log|S_\lambda|_+ + \frac{N - d_0}{2}\log\sigma^2,$$

where $N = nT$, $|\cdot|_+$ denotes the product of positive eigenvalues, and d_0 is the dimension of the unpenalized component after imposing the relevant identifiability constraints. The data-fit contribution enters through $Q_\lambda(\hat{\theta}_\lambda)$ and the scale term. The determinant terms adjust the marginal likelihood for the design, penalty, and dimension of the penalized and unpenalized coefficient spaces, thereby discouraging overly flexible smooths when the smoothing parameters are chosen. The `bam` option `fREML` is a computationally efficient REML-type implementation for large data sets (Wood, 2011; Wood et al., 2015, 2017). Thus, for the estimators used below, the coefficient objective is "RSS plus a quadratic wiggliness penalty," while REML/fREML selects the penalty weights instead of requiring the researcher to choose them by hand.

For the proposed method to approximate an unrestricted unknown function asymptotically, the basis dimension must satisfy $K_n \to \infty$ with the sample size, subject to appropriate regularity and rate conditions. This permits approximation error to vanish, whereas a fixed basis generally leads to a finite-dimensional projection limit and can leave nonvanishing approximation error. Cross-sectional penalized-spline theory studies the joint roles and relative rates of the sample size, basis dimension, and penalty, including regimes in which the estimator behaves more like a regression spline or a smoothing spline (Hall and Opsomer, 2005; Kauermann et al., 2009; Claeskens et al., 2009; Xiao, 2019; Huang and Su, 2021). In practice, a user may specify a basis dimension K_n that increases with sample size. Unlike conventional unpenalized series estimation, however, K_n does not by itself determine the complexity of the fitted function. It determines the available approximation space and places an upper bound on the attainable effective degrees of freedom. Conditional on that space, fREML estimates the smoothing parameter or parameters; these estimates, together with the design and penalty matrices, determine the effective degrees of freedom of the fitted smooth. Allowing K_n to grow therefore permits effective complexity to increase without requiring it to do so. With a fixed $K_n = K$, the estimated EDF may still increase with sample size as the selected penalty changes, but it remains bounded by the ceiling imposed by K.

The cluster-robust interpretation used here is most naturally associated with large-n, fixed-T panel asymptotics: the number of independent unit clusters grows, the number of time periods remains fixed, and arbitrary within-unit dependence is allowed.

I regard the finite-dimensional spline representation as a working approximation to the unknown function and do not assume that the population function lies exactly in the selected basis space. Consequently, fREML regularization controls complexity within the chosen space but does not eliminate basis-truncation bias. In conducting inference, I treat the working basis as sufficiently rich and do not formally account for truncation bias. To interpret the procedure as inference on an unrestricted population function under growing-basis asymptotics, one would additionally require the approximation bias to be negligible relative to the relevant sampling error.

In the simulations and accompanying code, I use $K = 20$ as the baseline. For supported smooths, `gam.check()` and `k.check()` provide the usual `mgcv` basis-dimension diagnostic: a low k-index, especially together with an EDF close to its attainable ceiling, can indicate residual pattern that a larger basis could capture (Wood, 2017). Its simulated p-value is obtained by randomly reshuffling observation-level residuals, however, and therefore does not preserve within-unit dependence. Moreover, the check does not support the matrix-valued covariate used by the first-difference linear-functional smooth. In panel applications I therefore treat the p-value as exploratory where it is available. My primary practical diagnostic is to increase K, for example by a factor of two, and examine both the stability of the substantive conclusions and the EDF relative to its attainable ceiling. The sensitivity analysis reported in Appendix Section A.2 evaluates this diagnostic. If the conclusions change materially or the EDF is close to its ceiling, K should be increased and the diagnostic repeated.

While REML/fREML is the default choice in `mgcv::bam`, one could use a more familiar generalized cross-validation (GCV) criterion. REML/fREML does not require a fully parametric likelihood model for the substantive econometric object of interest; instead, it is used as a data-driven rule for selecting the smoothing parameters. Other smoothing selection criteria, including GCV/Cp-type criteria, are also available in `mgcv`. I use fREML in the simulations because REML-type criteria are often more stable in finite samples and less prone to undersmoothing than GCV-type criteria (Reiss and Ogden, 2009). The coefficient and covariance construction can also be used after a GCV/Cp fit, but `bam`'s discrete fitting mode is specific to fREML, so switching to GCV/Cp also requires `discrete = FALSE`.

4 Inference

Standard `mgcv` returns a Bayesian or frequentist covariance matrix that assumes independent, homoskedastic observations. For panel data with within-unit dependence, a cluster-robust analog is needed. This section develops a penalty-adjusted cluster-robust covariance estimator from the fitted `bam` object.

Let R denote the fitted `mgcv` linear predictor matrix, S_λ the penalty matrix, and $\hat{u}$ the residual vector. For Gaussian identity-link models without observation weights, the `mgcv` working-weight matrix is the identity matrix, so the penalized Hessian matrix is

$$B_\lambda = R'R + S_\lambda. \tag{4.1}$$

A penalty-adjusted clustered sandwich then has the form

$$\widehat{V} = B_\lambda^{-1}\hat{\Omega}B_\lambda^{-1}, \quad \hat{\Omega} = \sum_{i=1}^{n} \hat{s}_i \hat{s}_i', \quad \hat{s}_i = \sum_{t=1}^{T} R_{it}' \hat{u}_{it}, \tag{4.2}$$

where R_{it} is the row of R for observation (i,t). Here, $\hat{\Omega}$ is the unit-clustered score covariance component: it first sums the score contributions $R_{it}'\hat{u}_{it}$ over time within each unit and then forms outer products of the unit-level sums. This construction has the same unit-level outer-product form as the robust covariance estimator for within-group estimators in Arellano (1987). The resulting sandwich estimator therefore allows arbitrary heteroskedasticity and serial dependence within units under independence across units. In $\widehat{V}$, the outer factors B_λ^{-1} incorporate the spline penalty. Thus, $\widehat{V}$ can be viewed as a penalty-adjusted spline analog of the cluster sandwich used in series-based semiparametric panel estimation. When $S_\lambda = 0$, $B_\lambda = R'R$, and $\widehat{V}$ reduces to the corresponding unpenalized series-style cluster sandwich.

4.1 Inference on Finite-Dimensional Components

For a scalar finite-dimensional parameter β, or for one component of a vector β, let $\hat{\sigma}_\beta^2$ denote the corresponding diagonal element of $\widehat{V}$. Under the Gaussian approximation,

$$\frac{\hat{\beta} - \beta}{\hat{\sigma}_\beta} \approx N(0, 1).$$

Thus an approximate $1 - \alpha$ confidence interval for β is

$$\left[\hat{\beta} - z_{1-\alpha/2}\hat{\sigma}_\beta,\ \hat{\beta} + z_{1-\alpha/2}\hat{\sigma}_\beta\right].$$

4.2 Inference on the Nonparametric Component

For inference on the nonparametric component, let $G = \{x_1, \ldots, x_J\}$ be the evaluation grid. Let R_G denote the grid `lpmatrix`, and let $R_G^{(g)}$ be the same matrix after all columns not belonging to the smooth of interest have been set to zero. Because the level of g is not separately identified from the intercept and unit effects, the target is the grid-centered smooth. Let

$$C_G = I_J - \mathbb{1}\mathbb{1}'/J$$

be the grid-centering matrix, where $\mathbb{1}$ is the J-vector of ones. Premultiplication by C_G subtracts the average over the J grid points. The matrix that maps fitted coefficients into centered smooth values is

$$L_G = C_G R_G^{(g)}.$$

Thus

$$\hat{g}_G^c = L_G\hat{\theta}, \qquad \hat{g}_G^c(x_j) = \hat{g}(x_j) - \frac{1}{J}\sum_{\ell=1}^{J} \hat{g}(x_\ell),$$

and the estimated covariance matrix for the centered grid vector is

$$\widehat{\Sigma}_G = L_G \widehat{V} L_G'.$$

The following Gaussian coverage approximations condition on the selected smoothing parameters and require smoothing and basis-approximation biases to be negligible relative to sampling uncertainty. Pointwise and uniform bands use the same estimated grid covariance matrix but answer different questions. A pointwise $1-\alpha$ interval at grid point x_j is

$$\hat{g}^c_G(x_j) \pm z_{1-\alpha/2}\hat{\sigma}_j, \qquad \hat{\sigma}^2_j = (\widehat{\Sigma}_G)_{jj}.$$

This interval is constructed separately at each x_j. For each individual point,

$$P\left(g^c_G(x_j) \in \left[\hat{g}^c_G(x_j) - z_{1-\alpha/2}\hat{\sigma}_j,\, \hat{g}^c_G(x_j) + z_{1-\alpha/2}\hat{\sigma}_j\right]\right) \approx 1-\alpha.$$

This pointwise statement does not imply that all intervals cover the curve simultaneously. The distinction between pointwise component intervals and whole-function coverage is standard in spline and GAM inference; see, for example, Marra and Wood (2012) on coverage of GAM component intervals.

A uniform band instead controls the probability that the entire reported grid curve is covered simultaneously:

$$P\left(g^c_G(x_j) \in [\hat{g}^c_G(x_j) - c_{1-\alpha}\hat{\sigma}_j,\, \hat{g}^c_G(x_j) + c_{1-\alpha}\hat{\sigma}_j] \text{ for all } j = 1,\dots,J\right) \approx 1-\alpha.$$

The critical value $c_{1-\alpha}$ is the $1-\alpha$ quantile of the maximum absolute standardized error over the grid. Each standardized component is marginally standard normal, but the maximum depends on their joint correlation structure. For a spline fit, nearby grid points are typically strongly correlated, so the maximum is neither the maximum of independent standard normals nor a single standard normal. Except in special cases, there is no simple scalar closed-form critical value; simultaneous bands for penalized splines are therefore commonly constructed from the approximating Gaussian distribution (Krivobokova et al., 2010). In the simulations I approximate this critical value from an auxiliary Gaussian vector

$Z_G \sim N(0, \widehat{\Sigma}_G)$, where Z_G is simulated rather than observed in the data:

$$c_{1-\alpha} = \text{quantile}_{1-\alpha}\left(\max_{1\leq j\leq J} |Z_{G,j}/\hat{\sigma}_j|\right).$$

This is a simultaneous band on the reported grid. Coverage is evaluated over the 50-point grid used in the simulation.

5 Implementation Guide

This section translates the preceding estimation and inference formulas into `mgcv` code. Suppose that one wants to estimate the model from equation 2.2, with scalar X_1 and X_2. There are several ways to implement the estimator, and the same post-estimation covariance calculations can then be applied to the fitted object.

5.1 Estimation

5.1.1 Fixed Effects

The main implementation uses factor fixed effects:

```
df$unit <- factor(df$unit)
bam(y ~ x1 + s(x2, k = 20) + unit,
    data = df,
    method = "fREML",
    select = TRUE,
    discrete = TRUE)
```

5.1.2 Penalized Unit Effects

An alternative regularized specification uses a penalized unit effect:

```
bam(y ~ x1 + s(x2, k = 20) + s(unit, bs = "re"),
    data = df,
```

```
    method = "fREML",
    select = TRUE,
    discrete = TRUE)
```

This specification is often useful for prediction or as a shrinkage comparison, but it changes the estimator when μ_i is correlated with the regressors. The fixed-effects interpretation in the usual panel-data sense comes from the factor-indicator or first-differenced specifications.

5.1.3 First Differences

Fixed effects can also be removed by first differencing. A constrained smooth of the form

$$g(X_{it}) - g(X_{i,t-1})$$

can be represented in `mgcv` using linear functional smooths. The implementation constructs the matrix of current and lagged covariates and the corresponding $+1/-1$ weights.

To construct the first-differenced data, suppose that the levels data frame `df` contains `unit`, an integer period index `time`, and the variables `y`, `x1`, and `x2`. I sort observations within units and retain only pairs one period apart. The resulting `df_fd` contains the differenced outcome and linear regressor, the current and lagged values of the smooth's covariate, and the unit identifier for subsequent clustering. The following code constructs this data frame and then fits the first-differenced model used in the simulations:

```
df <- df[order(df$unit, df$time), ]
df_fd <- do.call(rbind, lapply(split(df, df$unit), function(d) {
  now <- which(diff(d$time) == 1L) + 1L
  previous <- now - 1L
  data.frame(
    unit = d$unit[now], time = d$time[now],
    dy = d$y[now] - d$y[previous],
    dx1 = d$x1[now] - d$x1[previous],
    x2_now = d$x2[now], x2_lag = d$x2[previous]
  )
}))
X_pair <- cbind(df_fd$x2_now, df_fd$x2_lag)
L_pair <- cbind(rep(1, nrow(df_fd)), rep(-1, nrow(df_fd)))

```

```
fit <- bam(dy ~ dx1 + s(X_pair, by = L_pair, k = 20),
    data = list(
      dy = df_fd$dy,
      dx1 = df_fd$dx1,
      X_pair = X_pair,
      L_pair = L_pair
    ),
    method = "fREML",
    select = TRUE,
    discrete = TRUE)
```

The construction also accommodates unbalanced panels and gaps: it never differences across units or across nonconsecutive periods. Use `df_fd$unit` as the clustering vector, aligned with the rows of the differenced data.

Here `dy` and `dx1` are first differences, while `X_pair` stores $(X_{2,it}, X_{2,i,t-1})$ and `L_pair` stores the corresponding $(1, -1)$ weights. The term `s(X_pair, by = L_pair)` therefore estimates a single smooth $g(\cdot)$ evaluated at both arguments and constrained to enter as $g(X_{2,it}) - g(X_{2,i,t-1})$.

This transformation is also the starting point of Pütz and Kneib (2018) and their `pamfe` package. Their implementation uses penalized P-splines, a generalized least-squares correction for the moving-average dependence induced by differencing under their baseline error model, and volume-of-tube simultaneous bands. The workflow here instead represents the differenced function as a `bam` linear-functional smooth and applies the same penalty-adjusted unit-cluster sandwich and Gaussian sup-t construction used for the factor fixed-effects estimator. The latter permits heteroskedasticity and general within-unit dependence without specifying a parametric covariance model.

5.1.4 Additional Options

By default, `s()` smooths in `mgcv` use thin plate regression spline bases, specified by `bs = "tp"` (Wood, 2003). Researchers can also choose other spline bases, such as cubic regression splines with `bs = "cr"` or P-splines with `bs = "ps"`.

The same syntax extends naturally to varying-coefficient models. For example, a model in which the coefficient on $X_{1,it}$ varies smoothly with $X_{2,it}$,

$$Y_{it} = \beta_0(X_{2,it}) + X_{1,it}\beta_1(X_{2,it}) + \mu_i + \varepsilon_{it},$$

can be estimated using a by-variable smooth:

```
bam(y ~ s(x2, k = 20) + s(x2, by = x1, k = 20) + unit,
    data = df,
    method = "fREML",
    select = TRUE,
    discrete = TRUE)
```

The simulations below focus on one-dimensional smooths; the varying-coefficient and tensor-product examples are included to show how the same fixed-effects syntax extends to richer specifications.

For a fully nonparametric bivariate component $g(X_{1,it}, X_{2,it})$, one can use a tensor-product smooth:

```
bam(y ~ te(x1, x2, k = c(10, 10)) + unit,
    data = df,
    method = "fREML",
    select = TRUE,
    discrete = TRUE)
```

If the goal is to separate main effects from interactions, `ti()` provides the corresponding tensor-product interaction:

```
bam(y ~ s(x1, k = 10) + s(x2, k = 10) + ti(x1, x2, k = c(10, 10)) + unit,
    data = df,
    method = "fREML",
    select = TRUE,
    discrete = TRUE)
```

The argument `k` corresponds to the basis dimension K_n discussed in Section 3. If `k` is omitted, `mgcv` uses default basis dimensions. The functions `gam.check()` and `k.check()`

report the standard basis-dimension diagnostic for supported smooths, but their residual-reshuffling p-values do not preserve panel dependence, and the check is unavailable for the matrix-valued covariate used in the first-difference fit. I therefore use them as exploratory checks and recommend refitting at a larger `k` while examining the stability of the substantive conclusions and the fitted EDF.

For `bam`, the default smoothing parameter selection criterion is fREML, corresponding to `method = "fREML"`. Researchers who wish to use the GCV/Cp criterion instead can specify `method = "GCV.Cp"`. Because `bam`'s discrete fitting mode is available only with fREML, a GCV/Cp fit must also set `discrete = FALSE` or omit that argument; the discrete-workflow computation benchmarks below do not automatically carry over to GCV/Cp. The default choices are best viewed as convenient and computationally efficient options, rather than binding or unusual constraints imposed on the estimator.

5.2 Inference

After fitting any of the fixed-effects, penalized-unit-effects, or first-differenced specifications, standard errors are computed from the fitted `bam` object rather than from the default `mgcv` covariance matrix. The fitted linear predictor matrix is obtained with

```
R <- predict(fit, type = "lpmatrix")
u <- residuals(fit, type = "response")
```

and the unit-level score sums are formed by summing the row-wise products of the design matrix and residuals within unit:

```
score_g <- rowsum(R * as.numeric(u),
                  group = as.factor(unit_id),
                  reorder = FALSE)
Omega_hat <- crossprod(score_g)
```

This command produces $\hat{\Omega}$ from equation (4.2).

For the penalty-adjusted covariance estimator, one needs to obtain B_λ^{-1} from equation (4.1). In the Gaussian identity-link case this is obtained from the Bayesian covariance matrix returned by mgcv, rescaled by the estimated residual variance:

```
B_lambda_inv <- vcov(fit, freq = FALSE) / summary(fit)$scale
V_penalty <- B_lambda_inv %*% Omega_hat %*% B_lambda_inv
```

The simulations also apply the usual finite-sample cluster correction for a one-way cluster-robust covariance matrix (Cameron and Miller, 2015). In this panel setting, each unit is one cluster, so the number of clusters is n. Let N denote the total number of observations and let d denote the substantive degrees of freedom of the fitted model. The correction used below is

$$\frac{n}{n-1}\frac{N-1}{N-d}.$$

For a balanced panel, $N = nT$; for an unbalanced panel, N is simply the total number of observed unit-time cells. For the fixed-effects specification, d excludes nuisance unit indicators, matching the within-estimator convention rather than inflating the adjustment by counting every unit effect. In code, the correction is:

```
is_unit_effect_coef <- function(coef_names, unit_var) {
  startsWith(coef_names, unit_var) |
    startsWith(coef_names, paste0("s(", unit_var, ")."))
}

substantive_edf <- function(fit, unit_var) {
  coef_names <- names(fit$edf)
  keep <- !is_unit_effect_coef(coef_names, unit_var)
  sum(fit$edf[keep])
}

cluster_correction <- function(fit, score_g, R,
                               unit_var = "unit") {
  n_clusters <- nrow(score_g)
  nobs <- nrow(R)
```

```
  df_used <- substantive_edf(fit, unit_var)

  (n_clusters / (n_clusters - 1)) *
    ((nobs - 1) / max(nobs - df_used, 1))
}

corr <- cluster_correction(fit, score_g, R,
                           unit_var = "unit")
V_penalty <- corr * V_penalty
```

The `unit_var` argument makes the convention explicit and avoids relying on one hardcoded factor name. For example, the application below uses `unit_var = "firm_fe"` for its levels specifications.

For pointwise and uniform bands for $g(\cdot)$, the fitted smooth is evaluated on a grid by calling `predict(..., type = "lpmatrix")` on the grid data. As in Section 4, the full prediction matrix should not be used unchanged, because it includes the intercept, linear regressors, and unit effects. In code, the construction of $L_G = C_G R_G^{(g)}$ corresponds to selecting the columns for `s(x2)`, zeroing the remaining columns, and centering the selected grid rows:

```
grid_df <- data.frame(
  x1 = 0,
  x2 = x_grid,
  unit = factor(reference_unit, levels = levels(df$unit))
)
Rg <- predict(fit, newdata = grid_df, type = "lpmatrix")

smooth_id <- which(vapply(fit$smooth, `[[`, "", "label") == "s(x2)")
sm <- fit$smooth[[smooth_id[1]]]
smooth_cols <- sm$first.para:sm$last.para

L <- Rg
L[, -smooth_cols] <- 0
```

```
L <- sweep(L, 2, colMeans(L), "-")

g_hat <- as.numeric(L %*% coef(fit))
C_grid <- L %*% V_penalty %*% t(L)
```

Pointwise intervals only require the diagonal of `C_grid`:

```
se_grid <- sqrt(pmax(diag(C_grid), 0))
z_crit <- qnorm(0.975)

point_lower <- g_hat - z_crit * se_grid
point_upper <- g_hat + z_crit * se_grid
```

These intervals are appropriate for statements about a fixed grid point, such as the value of the centered smooth at $x = 0.5$. A plotted collection of pointwise intervals is visually useful, but it is not a simultaneous 95% band for the curve.

For a uniform band, first convert the covariance matrix into the correlation matrix of the standardized grid errors, simulate draws from the corresponding Gaussian approximation, and take the 95th percentile of the maximum absolute draw. The simulations below use 499 Gaussian draws for this step:

```
valid <- is.finite(se_grid) & se_grid > 1e-10
Corr <- C_grid[valid, valid, drop = FALSE] /
  outer(se_grid[valid], se_grid[valid])
Corr <- (Corr + t(Corr)) / 2

eig <- eigen(Corr, symmetric = TRUE)
vals <- pmax(eig$values, 0)
A <- eig$vectors %*% (sqrt(vals) * t(eig$vectors))

n_draws <- 499
sim <- matrix(rnorm(n_draws * length(vals)), nrow = n_draws) %*% t(A)
sup_crit <- max(z_crit,
  quantile(apply(abs(sim), 1, max), 0.95, names = FALSE)
```

```
)

uniform_lower <- g_hat - sup_crit * se_grid
uniform_upper <- g_hat + sup_crit * se_grid
```

The bands differ only in their critical values. The code shown above enforces $c_{0.95} \geq z_{0.975}$, since the 95th percentile of the maximum absolute standardized Gaussian error cannot be below the corresponding marginal cutoff; finite simulation can otherwise reverse their order for nearly linear smooths. More Gaussian draws, such as 999 or 1,999, reduce critical-value simulation error at additional computational cost. In the first-differenced implementation, the grid data evaluate the linear functional smooth with `X_pair = x_grid` and `L_pair = 1`; the same zeroing, centering, and pointwise-or-uniform critical-value steps then report the normalized level curve g_G^c.

6 Simulation Design

The simulation section studies both estimation accuracy and inference. For estimation of $g(\cdot)$, the benchmark loss is out-of-sample root mean squared error on an independently generated test set. For inference on β, the benchmark outcomes are empirical size under the null and empirical power under the alternative. For inference on $g(\cdot)$, the benchmark outcome is coverage of confidence bands.

For inference on $g(\cdot)$, I primarily use a penalty-adjusted cluster sandwich estimator. Appendix Section A.1 compares it with four alternative covariance estimators, including a series-style cluster sandwich estimator that ignores the penalty.

The simulation designs deliberately correlate the unit effects with the regressors. For each unit, draw a latent index $\alpha_i \sim \text{Beta}(2.2, 2.2)$ and set $\mu_i = 1.1(\alpha_i - \bar{\alpha}) + \nu_i$, where $\bar{\alpha}$ is the sample mean and $\nu_i \sim N(0, 1.25^2)$. In the one-regressor designs,

$$X_{it} = \min\{1, \max\{0, \alpha_i + \eta_{it}\}\}, \qquad \eta_{it} \overset{\text{iid}}{\sim} N(0, 0.18^2).$$

In the partially linear designs, $X_{2,it}$ is generated by the same equation and

$$X_{1,it} = 0.5X_{2,it} + 0.5\alpha_i + \zeta_{it}, \qquad \zeta_{it} \overset{\text{iid}}{\sim} N(0, 0.45^2).$$

The Gaussian noises are mutually independent before the common dependence through α_i. Thus the unobserved unit effect is correlated with the covariates. The independent test covariates used for Table 1 are generated from the same marginal law as X_{it}, using fresh draws of α and η.

The tables use a heteroskedastic Gaussian AR(1) error design as the primary specification. Let $r_{it} = 1 + 2\exp(0.75Z_{it})$ and define

$$\sigma_{it}^2 = 0.5^2 \frac{r_{it}}{N^{-1}\sum_{i,t} r_{it}}.$$

Conditional on the regressors, each unit's error vector is drawn independently as $\varepsilon_i \sim N(0, D_i\mathcal{R}_iD_i)$, where $D_i = \text{diag}(\sigma_{i1}, \ldots, \sigma_{iT})$ and $(\mathcal{R}_i)_{st} = \rho^{|s-t|}$ with $\rho = 0.5$. Hence the normalization sets the average conditional variance in each simulated panel to 0.5^2. Here $Z_{it} = X_{it}$ in the estimation-accuracy designs and $Z_{it} = X_{2,it}$ in the fully nonparametric coverage design. The partially linear outcome adds $X_{1,it}\beta$ and uses $Z_{it} = X_{1,it} + X_{2,it}$. Appendix Section A.1 also reports a heteroskedastic design with the same conditional variances and serially independent Gaussian errors. All reported tables use $M = 1{,}000$ Monte Carlo draws.

6.1 Estimation Accuracy

I consider three data generating processes (DGPs):

$$g_1(x) = 0.5 + 1.4x, \tag{6.1}$$

$$g_2(x) = 0.5 + 1.4x + 0.75(x - 0.5)^2, \tag{6.2}$$

$$g_3(x) = \sin(2\pi x), \tag{6.3}$$

and the outcome is then generated as

$$Y_{it} = g_j(X_{it}) + \mu_i + \varepsilon_{it}.$$

Table 1 reports estimation accuracy for the three sample-size designs. Each entry reports 100 times the root mean squared error for the centered function on an independently generated test set with 1,000 test points per draw. The fixed spline estimators use an unpenalized cubic B-spline basis constructed by `splines::bs()` with seven degrees of freedom, degree three, no intercept basis column, and boundary knots at zero and one. The levels estimator combines these basis columns with unit indicators; the first-difference estimator regresses the differenced outcome on current-minus-lagged basis rows without an intercept. This fixed-K benchmark can perform well when its number of series terms "matches" the smoothness of the unknown function, but is unable to adapt to the smoothness automatically. In the sine design, the fixed-K spline performs especially well, but it overfits in the other two designs and is dominated by the `bam` estimators.

Table 1: Out-of-sample RMSE for estimating g

DGP	Method	Mean RMSE ($\times$100)		
		$n = 200$, $T = 4$	$n = 200$, $T = 8$	$n = 500$, $T = 4$
linear	linear FE	1.035	0.772	0.641
linear	linear FD	0.982	0.680	0.632
linear	spline FE, fixed K	2.378	1.704	1.489
linear	spline FD, fixed K	2.247	1.504	1.433
linear	`bam` FD	1.182	0.807	0.757
linear	`bam` FE	1.186	0.882	0.744
linear	`bam` RE	1.213	0.884	0.795
quadratic	linear FE	5.985	5.906	5.892
quadratic	linear FD	5.986	5.895	5.888
quadratic	spline FE, fixed K	2.369	1.713	1.499
quadratic	spline FD, fixed K	2.255	1.511	1.425
quadratic	`bam` FD	1.713	1.204	1.136
quadratic	`bam` FE	1.767	1.320	1.169
quadratic	`bam` RE	1.781	1.325	1.211
$\sin(2\pi x)$	linear FE	48.076	47.996	47.951
$\sin(2\pi x)$	linear FD	48.122	48.015	47.968
$\sin(2\pi x)$	spline FE, fixed K	2.366	1.723	1.510
$\sin(2\pi x)$	spline FD, fixed K	2.270	1.541	1.450
$\sin(2\pi x)$	`bam` FD	2.625	1.904	1.807
$\sin(2\pi x)$	`bam` FE	2.678	2.052	1.834
$\sin(2\pi x)$	`bam` RE	2.692	2.056	1.862

Notes: Entries report 100 times the root mean squared error for the centered function on an independent test set. All designs use heteroskedastic AR(1) errors with $\rho = 0.5$.

6.2 Inference

The semiparametric inference simulations use the partially linear model

$$Y_{it} = X_{1,it}\beta + g(X_{2,it}) + \mu_i + \varepsilon_{it}.$$

The first exercise reports empirical rejection probabilities for tests of $H_0 : \beta = 1$, both under the null and under a fixed alternative.

Table 2 reports size results for tests of $H_0 : \beta = 1$ in the partially linear model. The design uses $g(x) = \sin(2\pi x)$ and the heteroskedastic AR(1) error process described above.

Table 2: Size results for tests of $H_0 : \beta = 1$

Method	Classic	Penalty	Mean $\hat{\beta}$	SD $\hat{\beta}$	Mean SE
Panel A: $n = 200$, $T = 4$					
`bam` FD	0.060	0.063	1.0003	0.0373	0.0372
`bam` FE	0.051	0.054	0.9999	0.0380	0.0385
`bam` RE	0.046	0.051	1.0037	0.0378	0.0381
Panel B: $n = 200$, $T = 8$					
`bam` FD	0.056	0.058	0.9994	0.0258	0.0250
`bam` FE	0.059	0.062	0.9992	0.0287	0.0278
`bam` RE	0.052	0.056	1.0012	0.0286	0.0277
Panel C: $n = 500$, $T = 4$					
`bam` FD	0.056	0.059	0.9999	0.0240	0.0236
`bam` FE	0.051	0.051	1.0006	0.0244	0.0244
`bam` RE	0.051	0.054	1.0043	0.0243	0.0242

Notes: The columns "Classic" and "Penalty" report empirical rejection frequencies using the cluster sandwich that ignores the penalty and the penalty-adjusted cluster sandwich, respectively. "Mean SE" reports the mean penalty-adjusted standard error.

Table 3 reports power results for the same test when the true value is $\beta = 1.075$.

Table 3: Power results for tests of $H_0 : \beta = 1$

Method	Classic	Penalty	Mean $\hat{\beta}$	SD $\hat{\beta}$	Mean SE
Panel A: $n = 200$, $T = 4$					
`bam` FD	0.515	0.525	1.0753	0.0373	0.0372
`bam` FE	0.478	0.485	1.0749	0.0380	0.0385
`bam` RE	0.525	0.545	1.0787	0.0378	0.0381
Panel B: $n = 200$, $T = 8$					
`bam` FD	0.829	0.832	1.0744	0.0258	0.0250
`bam` FE	0.756	0.758	1.0742	0.0287	0.0278
`bam` RE	0.772	0.779	1.0762	0.0286	0.0277
Panel C: $n = 500$, $T = 4$					
`bam` FD	0.872	0.873	1.0749	0.0240	0.0236
`bam` FE	0.870	0.872	1.0756	0.0244	0.0244
`bam` RE	0.909	0.919	1.0793	0.0243	0.0242

Notes: The columns "Classic" and "Penalty" report empirical rejection frequencies using the cluster sandwich that ignores the penalty and the penalty-adjusted cluster sandwich, respectively. The null is $H_0 : \beta = 1$, while the true value is $\beta = 1.075$. "Mean SE" reports the mean penalty-adjusted standard error.

Tables 4 and 5 report penalty-adjusted inference for the unknown function g. Table 4 continues with the partially linear model used for the β results, now focusing on the smooth component. Table 5 reports the corresponding results for the fully nonparametric model $Y_{it} = g(X_{it}) + \mu_i + \varepsilon_{it}$. In both cases, the target is the centered function evaluated on a grid of 50 points, with $g(x) = \sin(2\pi x)$. The "Avg. pointwise" column averages marginal coverage over the grid points, while "Min. pointwise" reports the worst grid-point coverage. The "Uniform" column reports the fraction of Monte Carlo draws in which the entire grid curve is covered simultaneously by the sup-t band, using 499 Gaussian draws to approximate the critical value in each fitted model. Appendix Tables A3 and A4 compare these results with the four alternative inference methods.

I generate fresh Gaussian draws in each Monte Carlo replication, share them across covariance methods and basis dimensions within that replication, and use separate random-number streams for data generation and critical-value simulation. The main tables, inference-method appendix, and $K = 20$ sensitivity results use the same underlying replications wherever their designs coincide.

Table 4: Penalty-adjusted coverage results for centered g: partially linear model

Method	Avg. pointwise	Min. pointwise	Uniform	Avg. width
Panel A: $n = 200$, $T = 4$				
`bam` FD	0.945	0.925	0.921	0.1785
`bam` FE	0.944	0.932	0.921	0.1818
`bam` RE	0.929	0.902	0.907	0.1782
Panel B: $n = 200$, $T = 8$				
`bam` FD	0.943	0.927	0.931	0.1294
`bam` FE	0.946	0.935	0.924	0.1387
`bam` RE	0.937	0.917	0.914	0.1376
Panel C: $n = 500$, $T = 4$				
`bam` FD	0.946	0.932	0.932	0.1231
`bam` FE	0.946	0.931	0.936	0.1250
`bam` RE	0.922	0.874	0.899	0.1229

Notes: Entries report coverage of penalty-adjusted confidence bands for the centered nonparametric component in the partially linear model. "Avg. width" is the average pointwise interval width.

Table 5: Penalty-adjusted coverage results for centered g: nonparametric model

Method	Avg. pointwise	Min. pointwise	Uniform	Avg. width
Panel A: $n = 200$, $T = 4$				
`bam` FD	0.941	0.929	0.914	0.1766
`bam` FE	0.938	0.919	0.918	0.1806
`bam` RE	0.923	0.898	0.880	0.1770
Panel B: $n = 200$, $T = 8$				
`bam` FD	0.945	0.925	0.937	0.1282
`bam` FE	0.945	0.926	0.926	0.1387
`bam` RE	0.936	0.918	0.908	0.1376
Panel C: $n = 500$, $T = 4$				
`bam` FD	0.947	0.935	0.931	0.1219
`bam` FE	0.945	0.930	0.939	0.1246
`bam` RE	0.922	0.886	0.887	0.1223

Notes: Entries report coverage of penalty-adjusted confidence bands for the centered nonparametric component. "Avg. width" is the average pointwise interval width.

Several findings are worth highlighting. For inference on β, the three `bam` methods yield similar finite-sample behavior. Under the null hypothesis, empirical rejection rates tend to lie between about 5% and 6%, close to the nominal 5% level. Inference on $g(\cdot)$ is more demanding. For factor fixed effects and first differences under the baseline penalty-adjusted cluster covariance, average pointwise coverage ranges from 0.938 to 0.947 in Tables 4 and 5, while every reported uniform-coverage estimate is below 0.95 and ranges from 0.914 to 0.939. Thus the pointwise shortfalls are small, but the uniform bands exhibit modest systematic undercoverage. Undercoverage is more pronounced for the penalized-unit-effect specification, sometimes falling below 0.90.

The appendix comparisons show how alternative covariance estimators change these results. For inference on the smooth, the cluster sandwich that ignores the penalty is markedly conservative. The observation-level `mgcv` sandwiches can materially under-cover for first-differenced models because they do not account for the dependence induced by differencing.

If undercoverage is a concern, especially for uniform bands, one could consider adding the heuristic expected-squared-bias inflation $\widehat{\Delta} = \widehat{V}^{B}_{\text{model}} - \widehat{V}^{F}_{\text{model}}$ to the baseline penalty-adjusted cluster covariance.[1] This difference between the model-based Bayesian and frequentist `mgcv` covariance matrices reduces undercoverage and brings coverage closer to nominal in some designs, especially for penalized unit effects (see also Nychka, 1988; Marra and Wood, 2012). It can, however, be too conservative: for factor fixed effects and first differences, the baseline covariance gives pointwise coverage closer to 95%, while the inflated covariance generally produces overcoverage for both pointwise and uniform bands. I therefore report inflation as a sensitivity alternative rather than an automatic correction. Appendix Section A.1 discusses this trade-off separately for pointwise and uniform bands. A wild cluster bootstrap that refits the `bam` model in each draw is another possible approach to inference (Cameron and Miller, 2015), but its performance for these penalized-spline panel estimators is beyond the scope of this paper.

Several mechanisms may contribute to the coverage distortions. When unit effects are correlated with regressors, shrinking them toward zero can introduce bias that the confidence bands do not account for. This is consistent with the weaker coverage of the penalized-unit-effect specification. Smoothing and finite-basis approximation can also contribute to estimator bias. A sandwich covariance describes sampling variability but does not correct the fitted function for such bias. In particular, REML/fREML selects smoothing parameters through a restricted-likelihood criterion, not a rule designed specifically for coverage-optimal undersmoothing or bias correction. Bias that is non-negligible relative to the standard error

[1] I thank the referee for suggesting this covariance adjustment.

can therefore impair coverage even when the covariance estimate is accurate (Marra and Wood, 2012). This issue is especially relevant to inference on $g(\cdot)$.

To assess the magnitude of estimator bias, I report a finite-replication Monte Carlo diagnostic in Appendix Section A.1. Within each design, I average the absolute Monte Carlo bias divided by the empirical Monte Carlo standard deviation over the 50 evaluation points. Across the two model classes, two error processes, and three sample-size configurations, this average ranges from 0.02 to 0.07 for first differences, from 0.03 to 0.08 for factor fixed effects, and from 0.22 to 0.51 for penalized unit effects. With $M = 1{,}000$, Monte Carlo sampling error alone produces an expected diagnostic value of approximately $\sqrt{2/(\pi M)} = 0.025$, even when the estimator's finite-sample bias is zero. Values near this reference level should therefore not be interpreted as evidence of nonzero bias. The much larger values for penalized unit effects remain informative and are consistent with their weaker coverage. The diagnostic does not isolate smoothing bias from other sources of bias, and a small grid average does not rule out localized bias that may matter for uniform coverage.

Finite-sample covariance estimation is another possible source of coverage distortion. Imbens and Kolesár (2016) and Pustejovsky and Tipton (2018) show that conventional heteroskedasticity- and cluster-robust procedures can have finite-sample distortions in various settings, motivating degrees-of-freedom and leverage-based corrections. Cameron and Miller (2015) and MacKinnon et al. (2023) emphasize that cluster-robust inference is asymptotic in the number of clusters, so modest over-rejection can remain in finite samples even when the number of clusters is not especially small. The penalized-spline setting adds a further source of finite-sample distortion because the covariance calculation treats the estimated smoothing parameter as fixed, ignoring the additional variability from its selection.

I assess sensitivity to the basis dimension by repeating all simulation designs with $K \in \{10, 20, 40\}$. Appendix Section A.2 reports the complete results, including average EDF and the frequency with which it reaches 90% or 95% of its attainable ceiling. Increasing K from 20 to 40 has little effect on RMSE, rejection frequencies, or coverage, and the EDF remains well below its ceiling. At $K = 10$, the sine estimation-accuracy designs reach the 90% threshold in essentially every draw and the 95% threshold in essentially every draw in the two larger configurations (Table A5). Binding is less pronounced in the inference designs: the 90% threshold is reached in some larger configurations, but the 95% frequency is zero throughout (Table A8). The lower sine-design RMSE sometimes obtained with $K = 10$ does not by itself justify a restrictive basis for population-function inference. Stability between $K = 20$ and $K = 40$ supports the baseline and increasing K as a practical check.

Finally, Appendix Table A9 reports a reproducible computation benchmark for the three `bam` estimators and the three sample sizes used in the simulations. At the largest design, $n = 500$ and $T = 4$, median fitting time is 0.072 seconds for first differences, 0.639 seconds for factor fixed effects, and 0.753 seconds for penalized unit effects; constructing the penalty-adjusted cluster covariance matrix takes 0.012, 0.337, and 0.591 seconds, respectively. Peak resident memory remains below 394 MB. The high-dimensional unit-effect coefficients, rather than the smooth alone, account for much of the additional fitting and covariance cost in the levels specifications.

The relative timing is implementation-specific. The factor fixed-effects specification includes unit indicators directly in the `bam` model matrix. Specialized packages such as `fixest` instead absorb high-dimensional fixed effects using fast demeaning algorithms (Bergé et al., 2026), but they are not integrated with `gam`/`bam`'s construction and penalization of smooths, joint fREML smoothing-parameter selection, and smooth-inference machinery.

7 Empirical Application

7.1 Data and Specifications

I illustrate the estimation and inference procedures using `EmplUK`, a public unbalanced panel distributed with the R package `plm` as `plm::EmplUK` (Croissant and Millo, 2008). The data contain 1,031 observations on 140 manufacturing firms in the United Kingdom from

1976 through 1984 and were used by Arellano and Bond (1991) in their application to employment equations. The variables used here are employment, wages, capital, and output. The industry-sector indicator is time invariant and is therefore absorbed by the firm effects.

The three levels specifications use all 1,031 observations. Each firm's observed years form a consecutive sequence, so first differencing yields 891 observations: one fewer observation for each of the 140 firms. Thus the reduction in sample size for the first-difference specification reflects the mechanical loss of each firm's first observed year rather than internal gaps in the panel.

Let ℓ_{it}, w_{it}, k_{it}, and y_{it} denote the logarithms of employment, wages, capital, and output, respectively, and define $X_{it} = (w_{it}, k_{it}, y_{it})'$. Because the panel is unbalanced, let $\mathcal{X}_i = (X_{i1}, \ldots, X_{iT_i})$ denote the complete observed covariate history for firm i. Throughout the application, I maintain the strict-exogeneity condition

$$E[\varepsilon_{it} \mid \mathcal{X}_i, \mu_i] = 0.$$

I first estimate the linear fixed-effects benchmark

$$\ell_{it} = \beta_w w_{it} + \beta_k k_{it} + \beta_y y_{it} + \mu_i + \tau_t + \varepsilon_{it}, \tag{7.1}$$

the partially linear specification

$$\ell_{it} = \beta_w w_{it} + \beta_k k_{it} + g_y(y_{it}) + \mu_i + \tau_t + \varepsilon_{it}, \tag{7.2}$$

and the additive specification

$$\ell_{it} = g_w(w_{it}) + g_k(k_{it}) + g_y(y_{it}) + \mu_i + \tau_t + \varepsilon_{it}. \tag{7.3}$$

These three levels models include firm effects μ_i and year effects τ_t. The partially linear model treats wages and capital linearly and allows the output relationship to be nonlinear, while the additive model is included as a more flexible diagnostic. I also estimate the first-difference counterpart of equation (7.2),

$$\Delta\ell_{it} = \beta_w \Delta w_{it} + \beta_k \Delta k_{it} + g_y(y_{it}) - g_y(y_{i,t-1}) + \delta_t + \Delta\varepsilon_{it}, \tag{7.4}$$

where $\delta_t = \tau_t - \tau_{t-1}$. Each smooth is represented by a cubic regression-spline basis with $K = 20$, and its smoothing parameter is selected by fREML.

The strict-exogeneity condition may be difficult to justify if wages, capital, and output are jointly determined with employment or respond to firm-level shocks; indeed, the original employment application emphasizes dynamics and potentially endogenous regressors. I therefore interpret the estimates as conditional relationships rather than causal or structural economic parameters. Addressing endogeneity would require appropriate instruments and, for the nonlinear components, potentially semiparametric or nonparametric instrumental-variable methods, which are beyond the scope of this paper. The application is intended as a worked illustration of the estimation and inference procedures on a publicly available panel rather than as a new structural analysis of UK labor demand.

7.2 Estimation and Inference

Starting from the logged variables in `empluk`, I construct the consecutive within-firm differences as follows:

```
empluk$firm_fe <- factor(empluk$firm)
empluk$year_fe <- factor(empluk$year)
empluk <- empluk[order(empluk$firm, empluk$year), ]
lag_by_firm <- function(x, firm)
  ave(x, firm, FUN = function(z) c(NA, head(z, -1)))
empluk$year_lag <- lag_by_firm(empluk$year, empluk$firm)
for (v in c("log_emp", "log_wage", "log_capital",
            "log_output")) {
  empluk[[paste0(v, "_lag")]] <-
    lag_by_firm(empluk[[v]], empluk$firm)
}
fd <- subset(empluk,
             !is.na(year_lag) & year - year_lag == 1)
fd_data <- with(fd, list(
```

```
  d_log_emp = log_emp - log_emp_lag,
  d_log_wage = log_wage - log_wage_lag,
  d_log_capital = log_capital - log_capital_lag,
  year_fd = factor(year),
  output_pair = cbind(log_output, log_output_lag),
  output_weights = outer(rep(1, nrow(fd)), c(1, -1)),
  firm = firm
))
```

The resulting `fd_data` object contains the 891 changes and current–lagged covariate pairs. The factor fixed-effects model uses all 1,031 rows of `empluk`, while the first-difference model uses `fd_data`. The two models are estimated as follows, where `firm_fe` and `year_fe` are factor variables:

```
empluk_pl <- bam(
  log_emp ~ log_wage + log_capital +
    s(log_output, bs = "cr", k = 20) +
    firm_fe + year_fe,
  data = empluk,
  method = "fREML", discrete = TRUE, select = TRUE
)
empluk_fd <- bam(
  d_log_emp ~ d_log_wage + d_log_capital + year_fd +
    s(output_pair, by = output_weights,
      bs = "cr", k = 20),
  data = fd_data,
  method = "fREML", discrete = TRUE, select = TRUE
)
```

The FD call uses the linear-functional smooth from Section 5. The `output_pair` object contains current and lagged log output, while `output_weights` contains the corresponding $+1/-1$ weights; together they produce $g_y(y_{it}) - g_y(y_{i,t-1})$. I apply the penalty-adjusted covariance estimator in Section 4 to both fits, clustering the score contributions by firm (using `fd_data$firm` for the first-difference model). The reported smooths are centered over evaluation grids covering the first through the 99th percentiles of the levels-sample covariate distributions. For the partially linear output functions, I report both pointwise and uniform 95% firm-clustered confidence bands. Each uniform critical value is obtained from 9,999 Gaussian draws using the estimated correlation structure of the centered smooth.

7.3 Results

Table 6 reports the linear coefficients and firm-clustered standard errors. Allowing output to enter nonlinearly in the levels model changes the wage and capital estimates only modestly. The partially linear factor fixed-effects estimates are -0.341 for wages and 0.545 for capital, while the first-difference estimates are -0.430 and 0.397, respectively. The levels output smooth has an estimated 2.87 effective degrees of freedom, indicating a relatively simple departure from linearity; the corresponding first-difference smooth has an EDF of 0.96 and is essentially linear. This near-linearity is selected by fREML within the spline space, not imposed by first differencing.

Table 6: Linear coefficients in the EmplUK application

Covariate	Linear FE	Partially linear FE	Partially linear FD
Log wages	-0.297	-0.341	-0.430
	(0.126)	(0.129)	(0.163)
Log capital	0.548	0.545	0.397
	(0.051)	(0.050)	(0.055)
Log output	0.265	—	—
	(0.153)		

Notes: The dependent variable is log employment in the levels specifications and its first difference in the FD specification. All specifications include year effects; the levels models also include firm effects. Standard errors clustered by firm are reported in parentheses. In the partially linear models, log output enters through a smooth function and therefore has no scalar coefficient; its estimated EDF is 2.87 in the factor fixed-effects specification and 0.96 in the first-difference specification.

Table 7: Comparison of specifications in the EmplUK application

Model	Wage EDF	Capital EDF	Output EDF	Model df	RMSE	AIC	BIC
Panel A: levels							
Linear FE	1.00	1.00	1.00	152.00	0.1180	-1177.4	-426.8
Partially linear FE	1.00	1.00	2.87	154.68	0.1172	-1185.5	-421.6
Additive GAM FE	8.84	11.61	2.51	175.68	0.1111	-1252.9	-385.3
Panel B: first differences							
Partially linear FD	1.00	1.00	0.96	12.00	0.1074	-1424.1	-1366.6

Notes: Linear components are assigned one degree of freedom. Model df is the effective parameter count returned by `logLik(fit)` and used in AIC and BIC; when available, it uses `mgcv`'s corrected EDF and therefore need not equal a simple sum of the displayed component EDFs. It includes year effects, the estimated scale parameter, and firm effects in the levels models. RMSE is calculated for log employment in Panel A and its first difference in Panel B. AIC and BIC are reported descriptively; fREML selects the smoothing parameters. RMSE, AIC, and BIC should not be compared across panels because the dependent variables, estimation samples, and objectives differ.

Figure 1 compares the centered covariate functions from the three levels specifications. The partially linear and additive models produce very similar output functions: employment rises with output over most of the observed range, but the estimated relationship flattens and then turns down at high output values. The capital function from the additive model remains close to the linear and partially linear fits. The most pronounced differences across specifications occur for wages, particularly near the sparsely populated boundaries of the covariate support.

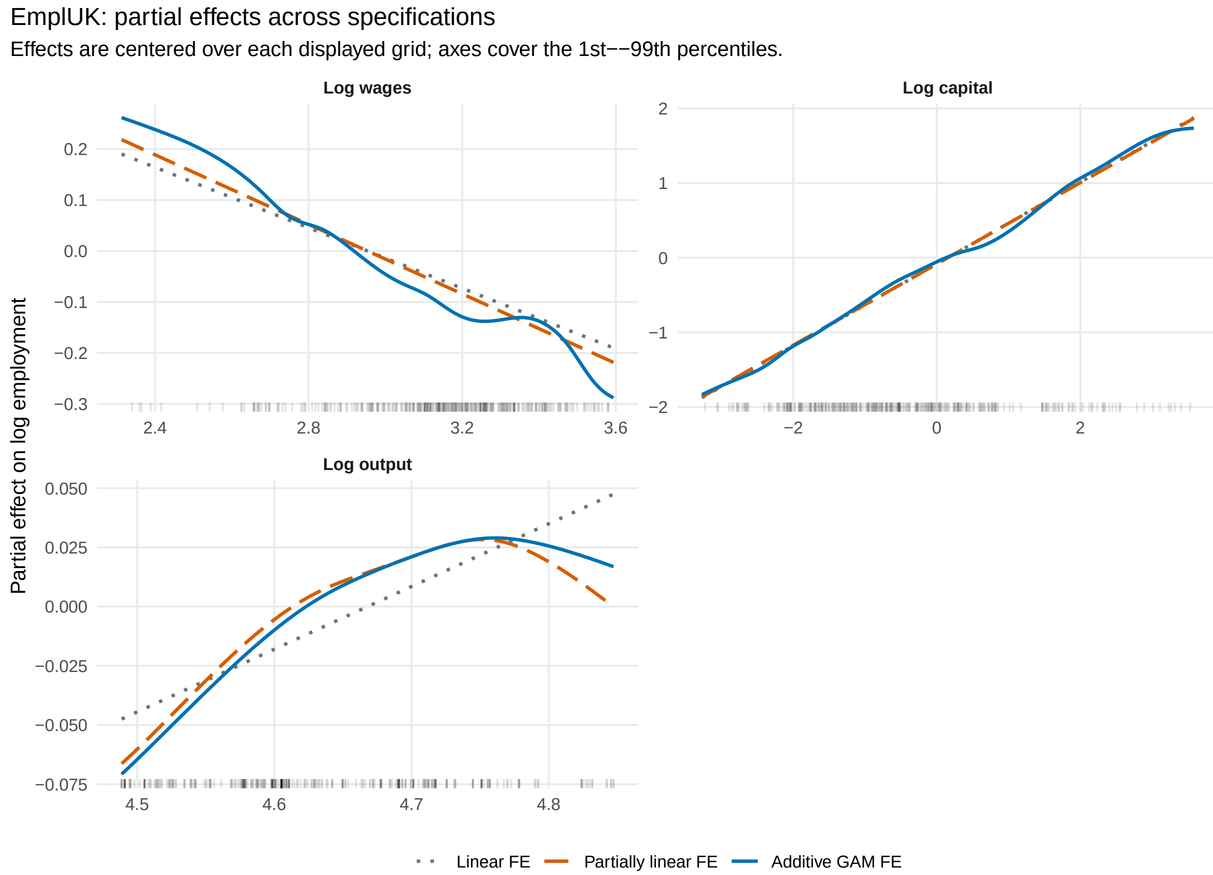

Figure 1: Estimated covariate functions in the `EmplUK` application
Notes: Each curve is centered over the displayed evaluation grid, which covers the first through the 99th percentiles of the corresponding logged covariate. All specifications include firm and year effects. The partially linear model treats wages and capital linearly and output nonlinearly; the additive model smooths all three covariates. Rugs show the observed covariate distributions.

Figure 2 reports the two partially linear output functions. The factor fixed-effects estimate turns down at the upper end of the support, while the first-difference estimate is nearly linear and increasing. Firm-clustered bands widen near the sparsely populated boundaries. The uniform band is wider for factor fixed effects; the two bands essentially coincide for the nearly linear first-difference fit.

Figure 2: Factor fixed-effects and first-difference estimates of the output function

Notes: The panels report the centered output functions from the partially linear factor fixed-effects and first-difference models. Shaded regions are 95% pointwise and uniform confidence bands constructed using the penalty-adjusted covariance estimator with clustering by firm. In each panel, the rug shows the distribution of log output in the corresponding estimation sample over the displayed first-through-99th-percentile range.

Although the factor fixed-effects and first-difference estimators perform similarly in the simulations, they produce markedly different estimates in this application. This contrast illustrates the practical trade-offs between the two approaches. The factor-indicator specification preserves the levels representation and uses all 1,031 observations, at the cost of enlarging the model matrix. First differencing removes the 140 firm effects before estimation and is computationally lighter, but yields 891 adjacent within-firm changes. Because the panel contains no internal gaps, the reduction is exactly one observation per firm. Differencing can magnify short-run measurement error, leave relatively little variation when regressors are highly persistent, and induce serial dependence in the transformed errors even when the level errors are serially independent. Both output estimates rise through the middle of the observed range, but the upper-tail downturn appears only in the factor fixed-effects estimate.

The contrasting shapes may reflect the different information and implicit weighting in the two penalized least-squares objectives. The levels objective is driven by within-firm deviations, whereas the unweighted first-difference objective is driven by adjacent changes. When those changes contain limited information about curvature, fREML may select a nearly linear function. In addition, the fREML fits use a working independent-error likelihood. The firm-cluster covariance adjustment applied after estimation affects inference but does not alter either the fitted function or the selected smoothing parameters. These features may help explain the lower first-difference EDF, although the application does not identify their separate contributions. The near-linear first-difference estimate should therefore not be interpreted as evidence that the population output effect is linear. I report both approaches as a sensitivity analysis rather than treat them as mechanically interchangeable.

The factor fixed-effects output estimate is insensitive to the working basis dimension: its EDF is 2.88, 2.87, and 2.87 at $K = 10$, 20, and 40, respectively, and its in-sample RMSE is unchanged to four decimal places. Among the levels specifications in Table 7, the fully additive model achieves the lowest in-sample RMSE and AIC but the highest BIC. Its wage and capital EDFs, as well as their detailed shapes, are more sensitive to K. I therefore treat the additive specification as a flexible diagnostic and use the two partially linear specifications as the main worked examples.

8 Conclusion

This paper provides a practical guide to estimating semiparametric and nonparametric panel data models with fixed effects using the `mgcv` package in R. The proposed approach connects the penalized sieve and spline literature in theoretical econometrics with REML/fREML implementations of penalized splines in statistics. The fixed-effects estimators use unit indicators or first differencing, while penalized unit effects provide a regularized alternative with a distinct shrinkage interpretation. The same syntax can be applied to partially linear, varying-coefficient, fully nonparametric, and related specifications.

The focus is implementation. I provide `mgcv::bam` commands for estimation and show

how to construct penalty-adjusted cluster-robust standard errors and confidence bands from the fitted model object. Across the one-dimensional linear, quadratic, and sinusoidal designs, a single penalized-spline specification performs reasonably without choosing a separate series dimension for each function, and tests for finite-dimensional parameters have empirical size close to nominal. For factor fixed effects and first differences, pointwise bands are near nominal, while uniform bands exhibit modest systematic undercoverage; the shortfalls are larger for penalized unit effects. Adding the model-based covariance inflation $\widehat{\Delta}$ generally reduces uniform-band undercoverage but can produce conservative bands, so I report it as a sensitivity alternative.

A direction for future work is to combine fast high-dimensional fixed-effect absorption with the penalized-smoothing, fREML selection, and inference features used here, thereby extending the implementation to larger panels.

Code Availability

The R code and replication materials for the simulations and empirical application are publicly available at `https://github.com/ikorolev89/mgcv-panel-data-replication`.

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Appendices

A Additional Simulation Results

A.1 Sensitivity to the Inference Method

Tables A1 and A2 repeat the size and power exercises using the five covariance estimators considered in the diagnostic analysis. Tables A3 and A4 report average pointwise and uniform coverage for the same estimators in the partially linear and fully nonparametric designs. Each table reports both the primary heteroskedastic AR(1) design and a heteroskedastic design with serially independent errors. The tables otherwise use the baseline basis dimension $K = 20$ and the simulation settings described in the main text.

For a given estimator and design, all five methods use the same penalized point estimates and residuals; only the covariance calculation changes. Let R be the fitted model matrix, $\hat{u}$ the fitted residuals, and $A_\lambda = (R'R + S_\lambda)^{-1}$. For the first-difference estimator, these objects refer to the transformed regression. In addition to the unit-clustered score covariance $\hat{\Omega}$ in equation (4.2), define its observation-level counterpart

$$\hat{\Omega}_{\text{obs}} = \sum_{i,t} R'_{it} R_{it} \hat{u}_{it}^2.$$

Unlike $\hat{\Omega}$, this expression omits cross-products between different observations from the same unit. The table headings correspond to the following covariance estimators:

1. *Classic.* The series-style unit-cluster sandwich ignores the penalty in the outer matrices:

$$\widehat{V}_{\text{Classic}} = c_0 (R'R)^{-1} \hat{\Omega} (R'R)^{-1}.$$

   If $R'R$ is singular because of redundant columns or identifying restrictions, $(R'R)^{-1}$ denotes the inverse on the full-rank, estimable component of the model matrix. In the code, the columns corresponding to the coefficients of interest are residualized without penalization on the nuisance columns, including unit effects where present, and a generalized inverse is used if the residualized cross-product remains singular. The residuals still come from the penalized fit: this is a covariance diagnostic, not a refitted unpenalized series estimator.

2. *Penalty.* The baseline is the penalty-adjusted unit-cluster sandwich from Section 4,

$$\widehat{V}_{\text{Penalty}} = c_\lambda A_\lambda \hat{\Omega} A_\lambda.$$

It retains the penalty in the outer matrices and allows heteroskedasticity and serial dependence within units.

3. *Penalty* $+\widehat{\Delta}$. This adds the difference between the ordinary model-based Bayesian and frequentist covariance matrices to the preceding clustered covariance:

$$\widehat{V}_{\text{Penalty}+\Delta} = \widehat{V}_{\text{Penalty}} + \widehat{\Delta}, \qquad \widehat{\Delta} = \widehat{V}^{B}_{\text{model}} - \widehat{V}^{F}_{\text{model}}.$$

Here, $\widehat{V}^{B}_{\text{model}}$ is returned by `vcov(fit, freq = FALSE)`, while $\widehat{V}^{F}_{\text{model}}$ uses `freq = TRUE`; both calls set `sandwich = FALSE`. In the Gaussian setup, their difference is $\hat{\sigma}^2 A_\lambda S_\lambda A_\lambda$. It represents an expected-squared-bias component under the smoothing prior, not an estimate of the realized frequentist bias under clustering (Wood, 2017; Marra and Wood, 2012). Adding it inflates uncertainty without changing or bias-correcting the fitted function.

4. *`mgcv`-F.* The frequentist observation-level sandwich is

$$\widehat{V}_{\text{mgcv-F}} = c_{\text{mgcv}} A_\lambda \hat{\Omega}_{\text{obs}} A_\lambda,$$

obtained with `vcov(fit, sandwich = TRUE, freq = TRUE)`. It allows observation-level heteroskedasticity but does not account for serial dependence, including dependence induced by first differencing.

5. *mgcv-B.* The Bayesian observation-level sandwich is defined directly as

$$\widehat{V}_{\text{mgcv-B}} := c_{\text{mgcv}} A_\lambda \hat{\Omega}_{\text{obs}} A_\lambda + \widehat{V}^B_{\text{model}} - \widehat{V}^F_{\text{model}} = \widehat{V}_{\text{mgcv-F}} + \widehat{\Delta}.$$

   It is returned by `vcov(fit, sandwich = TRUE, freq = FALSE)`. The final equality decomposes this matrix into the frequentist observation-level sandwich and the model-based covariance difference. Thus this column is a sandwich with bias inflation, not the ordinary model-based Bayesian covariance `vcov(fit, sandwich = FALSE, freq = FALSE)`, and it is not unit-clustered.

For completeness, the finite-sample multipliers are those used in the simulation code. With N regression observations and n units, the cluster methods use $c(d) = \{n/(n-1)\}\{(N-1)/(N-d)\}$. For c_λ, d is the fitted EDF excluding unit-effect coefficients; for c_0, it is the rank of the non-unit-effect columns of R. The `mgcv` methods instead use $c_{\text{mgcv}} = N/(N - d_{\text{all}})$, where d_{all} includes the EDF of the unit effects. These conventions are part of the methods being compared. All five calculations condition on the selected smoothing parameters. For smooth inference, each covariance is mapped to the same centered grid using $L_G \widehat{V} L_G'$, and the pointwise and simulated uniform critical values are constructed as in Section 4.

Table A1: Size results across inference methods for tests of $H_0 : \beta = 1$

Method	Classic	Penalty	Penalty $+\widehat{\Delta}$	`mgcv`-F	`mgcv`-B
Panel A: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 4$					
`bam` FD	0.060	0.063	0.063	0.081	0.081
`bam` FE	0.051	0.054	0.054	0.054	0.054
`bam` RE	0.046	0.051	0.051	0.056	0.055
Panel B: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 8$					
`bam` FD	0.056	0.058	0.058	0.087	0.087
`bam` FE	0.059	0.062	0.062	0.068	0.068
`bam` RE	0.052	0.056	0.056	0.064	0.064
Panel C: heteroskedastic AR(1), $\rho = 0.5$; $n = 500$, $T = 4$					
`bam` FD	0.056	0.059	0.059	0.080	0.080
`bam` FE	0.051	0.051	0.051	0.051	0.051
`bam` RE	0.051	0.054	0.051	0.057	0.056
Panel D: heteroskedastic serially independent errors; $n = 200$, $T = 4$					
`bam` FD	0.053	0.056	0.054	0.092	0.092
`bam` FE	0.050	0.050	0.050	0.048	0.048
`bam` RE	0.042	0.050	0.045	0.048	0.045
Panel E: heteroskedastic serially independent errors; $n = 200$, $T = 8$					
`bam` FD	0.055	0.055	0.055	0.102	0.102
`bam` FE	0.054	0.058	0.058	0.055	0.054
`bam` RE	0.058	0.060	0.060	0.059	0.058
Panel F: heteroskedastic serially independent errors; $n = 500$, $T = 4$					
`bam` FD	0.061	0.063	0.063	0.096	0.095
`bam` FE	0.059	0.061	0.061	0.060	0.060
`bam` RE	0.050	0.055	0.054	0.059	0.057

Notes: Entries are empirical rejection frequencies from 1,000 Monte Carlo replications. Classic is the unit-cluster sandwich without the penalty; Penalty includes it; and Penalty $+\widehat{\Delta}$ also adds the model-based Bayesian-minus-frequentist covariance difference, computed with `sandwich=FALSE`. The `mgcv`-F and `mgcv`-B columns use `vcov`($\widehat{m}$, `sandwich=TRUE`) with `freq=TRUE` and `freq=FALSE`, respectively; they aggregate scores by observation rather than by unit. The Monte Carlo standard error of a rejection frequency near 0.05 is approximately 0.007.

Table A2: Power results across inference methods for tests of $H_0 : \beta = 1$

Method	Classic	Penalty	Penalty $+\widehat{\Delta}$	`mgcv`-F	`mgcv`-B
Panel A: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 4$					
`bam` FD	0.515	0.525	0.524	0.597	0.596
`bam` FE	0.478	0.485	0.484	0.497	0.497
`bam` RE	0.525	0.545	0.538	0.549	0.542
Panel B: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 8$					
`bam` FD	0.829	0.832	0.832	0.878	0.877
`bam` FE	0.756	0.758	0.758	0.765	0.765
`bam` RE	0.772	0.779	0.778	0.794	0.793
Panel C: heteroskedastic AR(1), $\rho = 0.5$; $n = 500$, $T = 4$					
`bam` FD	0.872	0.873	0.873	0.916	0.916
`bam` FE	0.870	0.872	0.872	0.879	0.879
`bam` RE	0.909	0.919	0.916	0.923	0.922
Panel D: heteroskedastic serially independent errors; $n = 200$, $T = 4$					
`bam` FD	0.279	0.296	0.295	0.383	0.382
`bam` FE	0.338	0.354	0.353	0.353	0.352
`bam` RE	0.397	0.426	0.419	0.421	0.414
Panel E: heteroskedastic serially independent errors; $n = 200$, $T = 8$					
`bam` FD	0.522	0.526	0.526	0.646	0.645
`bam` FE	0.677	0.682	0.680	0.689	0.686
`bam` RE	0.703	0.714	0.712	0.717	0.716
Panel F: heteroskedastic serially independent errors; $n = 500$, $T = 4$					
`bam` FD	0.583	0.589	0.588	0.698	0.696
`bam` FE	0.703	0.706	0.705	0.706	0.706
`bam` RE	0.768	0.779	0.775	0.786	0.783

Notes: Entries are empirical rejection frequencies from 1,000 Monte Carlo replications. Classic is the unit-cluster sandwich without the penalty; Penalty includes it; and Penalty $+\widehat{\Delta}$ also adds the model-based Bayesian-minus-frequentist covariance difference, computed with `sandwich=FALSE`. The `mgcv`-F and `mgcv`-B columns use `vcov(`$\widehat{m}$`, sandwich=TRUE)` with `freq=TRUE` and `freq=FALSE`, respectively; they aggregate scores by observation rather than by unit. The Monte Carlo standard error is at most approximately 0.016.

Table A3: Coverage results across inference methods for centered g: partially linear model

	Classic		Penalty		Penalty $+\widehat{\Delta}$		`mgcv`-F		`mgcv`-B	
Method	Avg. point.	Uniform	Avg. point.	Uniform	Avg. point.	Uniform	Avg. point.	Uniform	Avg. point.	Uniform
Panel A: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 4$										
`bam` FD	0.996	0.999	0.945	0.921	0.969	0.979	0.927	0.888	0.960	0.967
`bam` FE	0.998	1.000	0.944	0.921	0.973	0.985	0.946	0.927	0.973	0.987
`bam` RE	0.997	1.000	0.929	0.907	0.963	0.980	0.933	0.910	0.964	0.986
Panel B: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 8$										
`bam` FD	0.994	0.997	0.943	0.931	0.966	0.976	0.919	0.854	0.951	0.951
`bam` FE	0.996	1.000	0.946	0.924	0.971	0.981	0.948	0.939	0.971	0.986
`bam` RE	0.995	1.000	0.937	0.914	0.966	0.978	0.939	0.924	0.968	0.983
Panel C: heteroskedastic AR(1), $\rho = 0.5$; $n = 500$, $T = 4$										
`bam` FD	0.994	1.000	0.946	0.932	0.968	0.977	0.927	0.888	0.957	0.968
`bam` FE	0.996	0.999	0.946	0.936	0.972	0.988	0.948	0.939	0.972	0.988
`bam` RE	0.993	0.997	0.922	0.899	0.958	0.972	0.924	0.905	0.959	0.969
Panel D: heteroskedastic serially independent errors; $n = 200$, $T = 4$										
`bam` FD	0.997	1.000	0.941	0.923	0.965	0.972	0.904	0.829	0.942	0.941
`bam` FE	0.997	1.000	0.946	0.932	0.972	0.980	0.947	0.940	0.973	0.981
`bam` RE	0.996	1.000	0.916	0.865	0.955	0.967	0.919	0.878	0.957	0.968
Panel E: heteroskedastic serially independent errors; $n = 200$, $T = 8$										
`bam` FD	0.994	0.998	0.942	0.903	0.961	0.967	0.895	0.782	0.933	0.910
`bam` FE	0.996	1.000	0.943	0.935	0.969	0.984	0.945	0.949	0.970	0.985
`bam` RE	0.995	0.999	0.932	0.904	0.963	0.978	0.935	0.922	0.964	0.979
Panel F: heteroskedastic serially independent errors; $n = 500$, $T = 4$										
`bam` FD	0.996	0.999	0.943	0.931	0.964	0.978	0.904	0.832	0.940	0.933
`bam` FE	0.997	0.999	0.945	0.940	0.971	0.987	0.946	0.940	0.971	0.986
`bam` RE	0.993	0.999	0.907	0.871	0.949	0.964	0.910	0.875	0.950	0.967

Notes: Entries report coverage probabilities based on 1,000 Monte Carlo replications. "Avg. point." averages coverage over the 50 evaluation points; "Uniform" is coverage of the entire curve by a sup-t band based on 499 Gaussian draws. The five inference methods are defined in the notes to Appendix Table A1. For single-point or uniform coverage near 0.95, the Monte Carlo standard error is approximately 0.007.

Table A4: Coverage results across inference methods for centered g: nonparametric model

	Classic		Penalty		Penalty $+\widehat{\Delta}$		`mgcv`-F		`mgcv`-B	
Method	Avg. point.	Uniform	Avg. point.	Uniform	Avg. point.	Uniform	Avg. point.	Uniform	Avg. point.	Uniform
Panel A: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 4$										
`bam` FD	0.996	1.000	0.941	0.914	0.965	0.972	0.923	0.872	0.956	0.957
`bam` FE	0.996	1.000	0.938	0.918	0.967	0.980	0.941	0.919	0.968	0.983
`bam` RE	0.995	0.999	0.923	0.880	0.960	0.971	0.925	0.900	0.961	0.974
Panel B: heteroskedastic AR(1), $\rho = 0.5$; $n = 200$, $T = 8$										
`bam` FD	0.994	1.000	0.945	0.937	0.966	0.980	0.922	0.885	0.953	0.965
`bam` FE	0.996	1.000	0.945	0.926	0.971	0.982	0.947	0.937	0.972	0.991
`bam` RE	0.995	1.000	0.936	0.908	0.966	0.981	0.938	0.918	0.967	0.987
Panel C: heteroskedastic AR(1), $\rho = 0.5$; $n = 500$, $T = 4$										
`bam` FD	0.994	1.000	0.947	0.931	0.967	0.980	0.929	0.889	0.958	0.960
`bam` FE	0.995	1.000	0.945	0.939	0.970	0.986	0.945	0.946	0.971	0.988
`bam` RE	0.993	0.999	0.922	0.887	0.956	0.970	0.923	0.892	0.957	0.968
Panel D: heteroskedastic serially independent errors; $n = 200$, $T = 4$										
`bam` FD	0.997	0.999	0.941	0.919	0.963	0.976	0.903	0.822	0.942	0.933
`bam` FE	0.998	1.000	0.941	0.918	0.969	0.980	0.944	0.929	0.970	0.983
`bam` RE	0.997	1.000	0.916	0.873	0.957	0.969	0.918	0.895	0.958	0.968
Panel E: heteroskedastic serially independent errors; $n = 200$, $T = 8$										
`bam` FD	0.995	1.000	0.942	0.918	0.961	0.971	0.889	0.774	0.930	0.912
`bam` FE	0.996	1.000	0.944	0.931	0.969	0.983	0.946	0.942	0.971	0.986
`bam` RE	0.995	1.000	0.933	0.902	0.963	0.977	0.935	0.914	0.964	0.979
Panel F: heteroskedastic serially independent errors; $n = 500$, $T = 4$										
`bam` FD	0.996	1.000	0.942	0.932	0.963	0.974	0.903	0.825	0.939	0.943
`bam` FE	0.997	1.000	0.944	0.930	0.970	0.985	0.945	0.932	0.971	0.986
`bam` RE	0.992	1.000	0.899	0.834	0.943	0.962	0.901	0.841	0.945	0.967

Notes: Entries report coverage probabilities based on 1,000 Monte Carlo replications. "Avg. point." averages coverage over the 50 evaluation points; "Uniform" is coverage of the entire curve by a sup-t band based on 499 Gaussian draws. The five inference methods are defined in the notes to Appendix Table A1. For single-point or uniform coverage near 0.95, the Monte Carlo standard error is approximately 0.007.

The coverage comparison does not uniformly favor either clustered penalty-based method. Averaging equally over the two model classes, two error processes, and three sample-size configurations, Penalty gives pointwise coverage of 0.943 and 0.944 for first differences and factor fixed effects, respectively, compared with 0.965 and 0.970 after adding $\widehat{\Delta}$. Uniform coverage rises from 0.924 to 0.975 for first differences and from 0.929 to 0.983 for factor fixed effects. Although the inflation brings some individual uniform-band results closer to 0.95, it generally overshoots nominal coverage for these two estimators. The gains are clearer for penalized unit effects: mean pointwise coverage rises from 0.923 to 0.958 and uniform coverage from 0.887 to 0.972. I therefore retain Penalty as the baseline while reporting Penalty $+\widehat{\Delta}$ as a useful sensitivity alternative, particularly when undercoverage is a concern. Its empirical gains do not establish a general bias correction or eliminate the consequences of shrinking correlated unit effects.

To assess estimator bias directly, I also use the grid-level Monte Carlo output underlying Tables A3 and A4. For each estimator and design, let $\widehat{b}(x_j)$ denote the Monte Carlo mean estimation error at grid point x_j, and let $\widehat{\mathrm{SD}}_{\mathrm{MC}}(x_j)$ denote the empirical Monte Carlo standard deviation of the estimation error. I summarize their relative magnitude by

$$\frac{1}{J}\sum_{j=1}^{J}\frac{|\widehat{b}(x_j)|}{\widehat{\mathrm{SD}}_{\mathrm{MC}}(x_j)}, \qquad J = 50.$$

Across the two model classes, two error processes, and three sample-size configurations, this diagnostic ranges from 0.024 to 0.069 for first differences, from 0.028 to 0.076 for factor fixed effects, and from 0.221 to 0.507 for penalized unit effects. Because changing the covariance estimator does not change the fitted function, the diagnostic is common across the five covariance methods compared in the tables. With $M = 1{,}000$, Monte Carlo sampling error alone produces an expected diagnostic value of approximately $\sqrt{2/(\pi M)} = 0.025$, even when the estimator's finite-sample bias is zero. Values near this reference level should therefore not be interpreted as evidence of nonzero bias. The substantially larger penalized-unit-effect values remain informative. The diagnostic measures total finite-sample bias and does not separately identify smoothing bias, finite-basis approximation bias, or bias induced by shrinking the unit effects.

A.2 Sensitivity to the Basis Dimension

This section repeats the estimation-accuracy and inference simulations at three basis dimensions: $K = 10$, $K = 20$, and $K = 40$. In addition to the performance measures reported in the main text, the tables report the mean fitted EDF and two measures of how frequently the basis dimension constrains the fitted smooth. The 90% and 95% binding measures are the fractions of Monte Carlo draws in which the fitted EDF is at least 90% and 95%, respectively, of its attainable ceiling. This ceiling is generally $K - 1$ for the centered levels smooths. It is also $K - 1$ for the first-difference linear-functional smooth because each current-minus-lagged basis row annihilates the constant basis direction.

Table A5: Sensitivity of function estimation to the basis dimension

		$K = 10$			$K = 20$			$K = 40$		
DGP	Method	RMSE	EDF/ceiling	Bind	RMSE	EDF/ceiling	Bind	RMSE	EDF/ceiling	Bind
Panel: $n = 200$, $T = 4$										
linear	`bam` FD	1.181	1.45/9	0.000/0.000	1.182	1.46/19	0.000/0.000	1.182	1.46/39	0.000/0.000
linear	`bam` FE	1.185	1.32/9	0.000/0.000	1.186	1.32/19	0.000/0.000	1.185	1.32/39	0.000/0.000
linear	`bam` RE	1.211	1.31/9	0.000/0.000	1.213	1.32/19	0.000/0.000	1.227	1.38/39	0.000/0.000
quadratic	`bam` FD	1.711	3.98/9	0.000/0.000	1.713	4.06/19	0.000/0.000	1.713	4.07/39	0.000/0.000
quadratic	`bam` FE	1.766	3.72/9	0.000/0.000	1.767	3.78/19	0.000/0.000	1.767	3.78/39	0.000/0.000
quadratic	`bam` RE	1.779	3.72/9	0.000/0.000	1.781	3.78/19	0.000/0.000	1.785	3.80/39	0.000/0.000
$\sin(2\pi x)$	`bam` FD	2.427	8.54/9	1.000/0.383	2.625	12.05/19	0.000/0.000	2.635	12.77/39	0.000/0.000
$\sin(2\pi x)$	`bam` FE	2.512	8.44/9	1.000/0.006	2.678	11.49/19	0.000/0.000	2.683	12.05/39	0.000/0.000
$\sin(2\pi x)$	`bam` RE	2.560	8.44/9	0.996/0.006	2.692	11.49/19	0.000/0.000	2.696	12.05/39	0.000/0.000
Panel: $n = 200$, $T = 8$										
linear	`bam` FD	0.805	1.48/9	0.000/0.000	0.807	1.50/19	0.000/0.000	0.807	1.50/39	0.000/0.000
linear	`bam` FE	0.879	1.31/9	0.000/0.000	0.882	1.33/19	0.000/0.000	0.882	1.33/39	0.000/0.000
linear	`bam` RE	0.881	1.31/9	0.000/0.000	0.884	1.33/19	0.000/0.000	0.885	1.35/39	0.000/0.000
quadratic	`bam` FD	1.201	4.69/9	0.000/0.000	1.204	4.85/19	0.000/0.000	1.204	4.86/39	0.000/0.000
quadratic	`bam` FE	1.319	4.24/9	0.000/0.000	1.320	4.33/19	0.000/0.000	1.320	4.34/39	0.000/0.000
quadratic	`bam` RE	1.325	4.24/9	0.000/0.000	1.325	4.33/19	0.000/0.000	1.326	4.34/39	0.000/0.000
$\sin(2\pi x)$	`bam` FD	1.681	8.78/9	1.000/1.000	1.904	13.70/19	0.000/0.000	1.917	15.06/39	0.000/0.000
$\sin(2\pi x)$	`bam` FE	1.864	8.67/9	1.000/1.000	2.052	12.74/19	0.000/0.000	2.058	13.64/39	0.000/0.000
$\sin(2\pi x)$	`bam` RE	1.935	8.66/9	0.992/0.992	2.056	12.74/19	0.000/0.000	2.062	13.64/39	0.000/0.000

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Table A5 (continued)

		$K = 10$			$K = 20$			$K = 40$		
DGP	Method	RMSE	EDF/ceiling	Bind	RMSE	EDF/ceiling	Bind	RMSE	EDF/ceiling	Bind
Panel: $n = 500$, $T = 4$										
linear	`bam` FD	0.757	1.48/9	0.000/0.000	0.757	1.49/19	0.000/0.000	0.758	1.49/39	0.000/0.000
linear	`bam` FE	0.744	1.34/9	0.000/0.000	0.744	1.34/19	0.000/0.000	0.744	1.34/39	0.000/0.000
linear	`bam` RE	0.795	1.33/9	0.000/0.000	0.795	1.34/19	0.000/0.000	0.797	1.34/39	0.000/0.000
quadratic	`bam` FD	1.134	4.73/9	0.000/0.000	1.136	4.90/19	0.000/0.000	1.136	4.91/39	0.000/0.000
quadratic	`bam` FE	1.169	4.45/9	0.000/0.000	1.169	4.56/19	0.000/0.000	1.169	4.57/39	0.000/0.000
quadratic	`bam` RE	1.211	4.45/9	0.000/0.000	1.211	4.56/19	0.000/0.000	1.213	4.59/39	0.000/0.000
$\sin(2\pi x)$	`bam` FD	1.584	8.79/9	1.000/1.000	1.807	13.82/19	0.000/0.000	1.820	15.22/39	0.000/0.000
$\sin(2\pi x)$	`bam` FE	1.637	8.74/9	1.000/1.000	1.834	13.26/19	0.000/0.000	1.841	14.34/39	0.000/0.000
$\sin(2\pi x)$	`bam` RE	1.812	8.72/9	0.983/0.983	1.862	13.26/19	0.000/0.000	1.869	14.35/39	0.000/0.000

Notes: RMSE entries report 100 times the root mean squared error. EDF/ceiling reports the mean fitted effective degrees of freedom and its attainable ceiling. Bind is reported as 90%/95%: the first number is the fraction of Monte Carlo draws in which EDF is at least 90% of the ceiling, and the second is the fraction in which EDF is at least 95% of the ceiling. The ceiling is $K - 1$ for the centered levels smooths and for the first-difference linear-functional smooth, whose contrast design annihilates the constant basis direction.

Table A6: Sensitivity of inference on β to the basis dimension

	Size			Power		
Method	$K = 10$	$K = 20$	$K = 40$	$K = 10$	$K = 20$	$K = 40$
Panel: $n = 200$, $T = 4$						
`bam` FD	0.063	0.063	0.063	0.528	0.525	0.525
`bam` FE	0.054	0.054	0.054	0.485	0.485	0.486
`bam` RE	0.053	0.051	0.051	0.539	0.545	0.545
Panel: $n = 200$, $T = 8$						
`bam` FD	0.058	0.058	0.059	0.833	0.832	0.830
`bam` FE	0.062	0.062	0.062	0.759	0.758	0.758
`bam` RE	0.057	0.056	0.056	0.779	0.779	0.779
Panel: $n = 500$, $T = 4$						
`bam` FD	0.058	0.059	0.058	0.877	0.873	0.874
`bam` FE	0.047	0.051	0.052	0.872	0.872	0.871
`bam` RE	0.055	0.054	0.054	0.919	0.919	0.919

Notes: Entries report rejection frequencies based on the penalty-adjusted cluster sandwich. Size uses $\beta = 1$ and tests $H_0 : \beta = 1$. Power uses $\beta = 1.075$ and tests the same null.

Table A7: Sensitivity of coverage for centered g to the basis dimension

		$K=10$			$K=20$			$K=40$		
Model	Method	Avg. point	Min. point	Uniform	Avg. point	Min. point	Uniform	Avg. point	Min. point	Uniform
Panel: $n=200$, $T=4$										
PL	`bam` FD	0.946	0.929	0.929	0.945	0.925	0.921	0.944	0.927	0.921
PL	`bam` FE	0.946	0.934	0.926	0.944	0.932	0.921	0.944	0.932	0.919
PL	`bam` RE	0.930	0.900	0.898	0.929	0.902	0.907	0.929	0.902	0.906
NP	`bam` FD	0.942	0.929	0.918	0.941	0.929	0.914	0.941	0.931	0.910
NP	`bam` FE	0.938	0.921	0.916	0.938	0.919	0.918	0.937	0.917	0.916
NP	`bam` RE	0.921	0.892	0.880	0.923	0.898	0.880	0.922	0.899	0.879
Panel: $n=200$, $T=8$										
PL	`bam` FD	0.944	0.930	0.932	0.943	0.927	0.931	0.943	0.926	0.931
PL	`bam` FE	0.945	0.935	0.935	0.946	0.935	0.924	0.946	0.934	0.923
PL	`bam` RE	0.937	0.904	0.915	0.937	0.917	0.914	0.937	0.916	0.916
NP	`bam` FD	0.948	0.928	0.927	0.945	0.925	0.937	0.944	0.924	0.933
NP	`bam` FE	0.946	0.931	0.930	0.945	0.926	0.926	0.944	0.926	0.929
NP	`bam` RE	0.937	0.915	0.909	0.936	0.918	0.908	0.936	0.917	0.909
Panel: $n=500$, $T=4$										
PL	`bam` FD	0.948	0.934	0.938	0.946	0.932	0.932	0.945	0.931	0.934
PL	`bam` FE	0.948	0.934	0.946	0.946	0.931	0.936	0.946	0.934	0.938
PL	`bam` RE	0.923	0.874	0.896	0.922	0.874	0.899	0.923	0.872	0.901
NP	`bam` FD	0.947	0.926	0.940	0.947	0.935	0.931	0.947	0.936	0.935
NP	`bam` FE	0.946	0.931	0.940	0.945	0.930	0.939	0.945	0.929	0.933
NP	`bam` RE	0.920	0.876	0.883	0.922	0.886	0.887	0.922	0.886	0.890

Notes: Entries report average pointwise, minimum pointwise, and uniform coverage for penalty-adjusted bands; PL and NP denote partially linear and nonparametric models.

Table A8: EDF diagnostics for the inference sensitivity simulations

		$K = 10$		$K = 20$		$K = 40$	
Model	Method	EDF/ceiling	Bind	EDF/ceiling	Bind	EDF/ceiling	Bind
Panel: $n = 200$, $T = 4$							
β	`bam` FD	7.79/9	0.009/0.000	9.55/19	0.000/0.000	9.81/39	0.000/0.000
β	`bam` FE	7.59/9	0.000/0.000	9.04/19	0.000/0.000	9.23/39	0.000/0.000
β	`bam` RE	7.60/9	0.000/0.000	9.05/19	0.000/0.000	9.24/39	0.000/0.000
PL g	`bam` FD	7.79/9	0.002/0.000	9.53/19	0.000/0.000	9.79/39	0.000/0.000
PL g	`bam` FE	7.59/9	0.000/0.000	9.04/19	0.000/0.000	9.23/39	0.000/0.000
PL g	`bam` RE	7.60/9	0.000/0.000	9.05/19	0.000/0.000	9.25/39	0.000/0.000
NP g	`bam` FD	7.81/9	0.009/0.000	9.60/19	0.000/0.000	9.86/39	0.000/0.000
NP g	`bam` FE	7.61/9	0.002/0.000	9.07/19	0.000/0.000	9.26/39	0.000/0.000
NP g	`bam` RE	7.62/9	0.003/0.000	9.07/19	0.000/0.000	9.27/39	0.000/0.000
Panel: $n = 200$, $T = 8$							
β	`bam` FD	8.31/9	1.000/0.000	11.07/19	0.000/0.000	11.57/39	0.000/0.000
β	`bam` FE	8.06/9	0.285/0.000	10.16/19	0.000/0.000	10.48/39	0.000/0.000
β	`bam` RE	8.06/9	0.284/0.000	10.16/19	0.000/0.000	10.48/39	0.000/0.000
PL g	`bam` FD	8.31/9	1.000/0.000	11.08/19	0.000/0.000	11.58/39	0.000/0.000
PL g	`bam` FE	8.06/9	0.261/0.000	10.18/19	0.000/0.000	10.51/39	0.000/0.000
PL g	`bam` RE	8.06/9	0.263/0.000	10.18/19	0.000/0.000	10.51/39	0.000/0.000
NP g	`bam` FD	8.32/9	1.000/0.000	11.10/19	0.000/0.000	11.61/39	0.000/0.000
NP g	`bam` FE	8.07/9	0.286/0.000	10.18/19	0.000/0.000	10.51/39	0.000/0.000
NP g	`bam` RE	8.07/9	0.294/0.000	10.19/19	0.000/0.000	10.51/39	0.000/0.000

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Table A8 (continued)

Model	Method	$K = 10$		$K = 20$		$K = 40$	
		EDF/ceiling	Bind	EDF/ceiling	Bind	EDF/ceiling	Bind
Panel: $n = 500$, $T = 4$							
β	`bam` FD	8.34/9	1.000/0.000	11.16/19	0.000/0.000	11.68/39	0.000/0.000
β	`bam` FE	8.21/9	0.988/0.000	10.63/19	0.000/0.000	11.02/39	0.000/0.000
β	`bam` RE	8.21/9	0.990/0.000	10.64/19	0.000/0.000	11.03/39	0.000/0.000
PL g	`bam` FD	8.34/9	1.000/0.000	11.19/19	0.000/0.000	11.71/39	0.000/0.000
PL g	`bam` FE	8.21/9	0.985/0.000	10.64/19	0.000/0.000	11.04/39	0.000/0.000
PL g	`bam` RE	8.21/9	0.986/0.000	10.65/19	0.000/0.000	11.05/39	0.000/0.000
NP g	`bam` FD	8.35/9	1.000/0.000	11.21/19	0.000/0.000	11.73/39	0.000/0.000
NP g	`bam` FE	8.22/9	0.989/0.000	10.65/19	0.000/0.000	11.05/39	0.000/0.000
NP g	`bam` RE	8.22/9	0.988/0.000	10.66/19	0.000/0.000	11.06/39	0.000/0.000

Notes: EDF/ceiling reports the mean fitted effective degrees of freedom and its attainable ceiling. Bind is reported as 90%/95%: the first number is the fraction of draws in which EDF is at least 90% of the ceiling, and the second is the fraction in which EDF is at least 95% of the ceiling. The β rows use the null design; its EDF diagnostics are numerically indistinguishable from those under the fixed alternative.

A.3 Computational Benchmarks

Table A9 reports computation time and peak resident memory for the three `bam` estimators at the sample sizes used in the simulations. The benchmark separates fitting time from the additional time required to construct the penalty-adjusted unit-cluster covariance matrix.

Table A9: Computation time and memory use

Method	Median fit time (s)	Median covariance time (s)	Peak RSS (MB)
Panel: $n = 200$, $T = 4$ *(800 levels; 600 FD observations)*			
`bam` FD	0.040	0.009	281.6
`bam` FE	0.068	0.035	274.7
`bam` RE	0.081	0.052	265.7
Panel: $n = 200$, $T = 8$ *(1,600 levels; 1,400 FD observations)*			
`bam` FD	0.041	0.010	269.4
`bam` FE	0.070	0.036	262.3
`bam` RE	0.083	0.053	268.4
Panel: $n = 500$, $T = 4$ *(2,000 levels; 1,500 FD observations)*			
`bam` FD	0.072	0.012	293.2
`bam` FE	0.639	0.337	380.5
`bam` RE	0.753	0.591	393.5

Notes: Each entry is based on 10 independently simulated panels from the $g(x) = \sin(2\pi x)$ heteroskedastic AR(1) design with $\rho = 0.5$ and $K = 20$. Fit and covariance columns report median elapsed (wall-clock) time across the 10 draws. Covariance time covers construction of the penalty-adjusted unit-cluster covariance matrix. Peak RSS is the maximum resident memory of the fresh R process that performs the 10 repetitions and therefore includes the R session and loaded packages. Benchmarks use a single R process with default threading. The factor-FE column uses direct unit indicators in `bam` and does not benchmark specialized high-dimensional fixed-effect absorption routines. Environment: Apple M5 Max, 18 cores (6 Super and 12 Performance), 48 GB, macOS 26.5.2, R version 4.6.0 (2026-04-24), `mgcv` 1.9-4.